

# TheAshantiHub — Project Scope & Timeline

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## 1. Executive Overview

TheAshantiHub currently exists as a **front-end-only prototype**: a single-page React application with a polished, Ghanaian-themed UI covering a business marketplace, admin dashboard, business dashboard, payments dashboard, and a credit-scoring dashboard. None of it is backed by a real system — every listing, transaction, credit score, and user is hardcoded mock data living in one 3,600-line file, and there is no server, database, authentication, or payment processing behind it.

This document scopes the work to turn that prototype into a real, working product: what exists today, what has to be built, in what order, on what stack, and on what timeline — including a compressed **Week 1 sprint** to get a mock-data-free MVP in front of the client, followed by a phased roadmap to bring payments, AI-assisted messaging, and enriched credit scoring online.

### Headline decisions:

- Backend: **Django + Django REST Framework**, PostgreSQL, Django Channels (later phase)
- Payments: **Hubtel** (Mobile Money + cards) — Phase 2
- AI: **Anthropic Claude API** for concierge search + business auto-reply — Phase 3
- Hosting: **Cloud VPS** (Docker Compose, mid-tier, ~\$50-85/mo) — chosen for cost efficiency over managed PaaS
- Mobile app: out of scope for this document; planned as a separate phase/spec after the web platform stabilizes

## 2. Current Implementation (As-Is)

| Area                      | Current state                                                                                                                                                                       |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Frontend                  | React 18 + Vite, no router — navigation via local <code>useState</code> flags ( <code>page</code> , <code>isAdmin</code> , <code>showBizDash</code> , etc.) in <code>App.jsx</code> |
| Backend                   | <b>None.</b> No server, no API, no database                                                                                                                                         |
| Data                      | Hardcoded in-file: <code>CATEGORIES</code> , <code>LISTINGS</code> , <code>MOCK_CREDIT_BUSINESSES</code> , <code>LENDING_PARTNERS</code> , <code>SCORE_FACTORS</code>               |
| Auth                      | Not implemented — <code>authModal</code> / <code>user</code> state exists and gates features, but no modal renders and signup/login no-ops                                          |
| Messaging                 | No in-app messaging; contact is <code>wa.me</code> deep links only ( <code>handleWA</code> )                                                                                        |
| Payments                  | <code>PaymentDashboard</code> renders against mock data only; no processor integrated                                                                                               |
| Credit scoring            | <code>CreditDashboard</code> / <code>ScoreGauge</code> compute against <code>MOCK_CREDIT_BUSINESSES</code> ; no real signals                                                        |
| Admin/Business dashboards | Full UI exists ( <code>AdminDashboard</code> , <code>BusinessDashboard</code> ), fed entirely by mock data; <code>isAdmin</code> toggled by a hidden 5-click gesture on the logo    |

|           |                                                                                            |
|-----------|--------------------------------------------------------------------------------------------|
| Analytics | In-memory event array ( <code>Analytics.track</code> ) — does not persist or send anywhere |
| Hosting   | Static Vercel deployment (SPA rewrite + security headers), no backend to host              |

**Gap in one sentence:** everything the UI *shows* exists; nothing it *depends on* is real.

### 3. Target Architecture

| Layer               | Choice                                                                                                                        | Rationale                                                                                                    |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Backend framework   | Django + Django REST Framework                                                                                                | Batteries-included (auth, admin, ORM, migrations), fast to build a solid API solo                            |
| Database            | PostgreSQL                                                                                                                    | Django's native production DB; handles relational data (users, listings, bookings, messages, scores) cleanly |
| Real-time messaging | Django Channels + Redis (Phase 3)                                                                                             | Needed once AI auto-reply/concierge requires live chat; deferred past Week 1 to control scope                |
| Auth                | Phone number + OTP (primary) and email + password (fallback)                                                                  | Matches how Ghanaian users already authenticate with Mobile Money; email as a fallback channel               |
| Payments            | Hubtel (Mobile Money + card aggregator)                                                                                       | Ghana-market fit, avoids direct telco integration complexity                                                 |
| AI                  | Anthropic Claude API                                                                                                          | Concierge (conversational listing discovery) + auto-reply (covers businesses when offline)                   |
| Media storage       | VPS local disk (Docker volume) + scheduled off-site backup                                                                    | Zero extra monthly cost; sufficient at launch scale                                                          |
| Frontend            | React 19 / TypeScript / Vite SPA, retained                                                                                    | Replace mock arrays/state with real API calls; no framework rewrite                                          |
| Hosting             | Single Cloud VPS, mid-tier (~\$50-80/mo)                                                                                      | Chosen for cost efficiency per client preference                                                             |
| Deployment          | Docker Compose (Django+Gunicorn, Postgres, Nginx+Certbot) + GitHub Actions (build & SSH deploy on push to <code>main</code> ) | Full control, no extra managed-PaaS layer, industry-standard, low ongoing cost                               |

### 4. Feature Scope

All mock data is being removed. Full platform scope, delivered in phases:

- **Week 1 MVP:** Real auth, real business/listing data, self-serve business signup with admin approval, simple in-app messaging, credit scoring v1 (non-payment signals), dashboards wired to live data
- **Phase 2:** Hubtel payments integration
- **Phase 3:** Real-time messaging upgrade (Channels/Redis) + AI concierge and auto-reply (Claude API)
- **Phase 4:** Credit scoring enriched with real payment/transaction history
- **Phase 5:** DevOps hardening — security review, monitoring, backups, load testing
- **Future (separate spec):** Mobile app

WhatsApp remains a simple `wa.me` deep-link handoff throughout (no backend integration, no AI on that channel) — this was an explicit scope decision to avoid WhatsApp Business Platform cost/complexity.

## 5. Week 1 Sprint Plan — "No More Mock Data" MVP

**Pace:** 16 hrs/day × 7 days ≈ 112 hours, solo developer, zero schedule slack.

| Day | Focus                                                                                                                                                                                                                                                                                                                                                                                              | Output                                                                                                  |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| 1   | Django + DRF + Postgres scaffold, Docker Compose dev environment, <code>User</code> model (phone + email), JWT auth skeleton                                                                                                                                                                                                                                                                       | Backend boots, migrations run, empty API live                                                           |
| 2   | Phone OTP (SMS via Hubtel SMS or Africa's Talking) + email/password auth flows                                                                                                                                                                                                                                                                                                                     | Real signup/login replaces the dormant <code>authModal</code> no-op                                     |
| 3   | <code>Business</code> & <code>Listing</code> models; migrate <code>CATEGORIES</code> / <code>LISTINGS</code> structure into Postgres; image upload (local disk); self-serve business signup API                                                                                                                                                                                                    | Mock listing data fully retired from the codebase                                                       |
| 4   | Admin approval queue wired into <code>AdminDashboard</code> ; frontend marketplace switched from mock arrays to real API calls; search/filter against real data                                                                                                                                                                                                                                    | Admin can approve/reject real business submissions                                                      |
| 5   | Simple polling-based in-app inbox (replaces <code>showMessaging</code> mock); reviews/ratings persisted to DB                                                                                                                                                                                                                                                                                      | Buyers and businesses can exchange real messages                                                        |
| 6   | Credit scoring v1 (listing completeness, response rate/time, review ratings, verification status, account age — <b>explicitly no payment history yet</b> ); <code>BusinessDashboard</code> / <code>CreditDashboard</code> / <code>PaymentDashboard</code> wired to live data ( <code>PaymentDashboard</code> shows ledger structure with a "Hubtel integration — Phase 2" state, not fake numbers) | All four dashboards render real data; credit score UI clearly labels which factors are live vs. pending |
| 7   | Production deploy (Docker Compose, Nginx, Certbot) to VPS; end-to-end QA; bug-fix buffer; client demo prep                                                                                                                                                                                                                                                                                         | Live, mock-data-free MVP on a real URL                                                                  |

**Explicit exclusions from Week 1 (confirmed, moved to Phase 2+):** Hubtel payment processing, mobile app, real-time chat, AI concierge/auto-reply, payment-based credit signals.

### Risks called out in advance:

- **SMS/OTP provider onboarding** (Hubtel SMS or Africa's Talking account approval) is outside developer control and is the single most likely thing to slip the schedule.
- **Zero slack.** This plan assumes no major blockers; any one surfacing (provider approval delay, VPS/DNS propagation issues, scope creep from the client mid-week) pushes the demo date.
- **Sixteen-hour days are not sustainable past this sprint.** Recommend returning to a normal pace for Phase 2 onward to avoid burnout-driven quality drop.

## 6. Phase 2+ Roadmap

Assuming a return to a sustainable full-time pace (~8 hrs/day) after the Week 1 crunch:

| Phase                                   | Scope                                                                                                                                                     | Est. duration          |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| 2 — Payments                            | Hubtel integration: checkout initiation, webhook/callback handling, transaction ledger, receipts                                                          | 2-3 weeks              |
| 3 — Real-time + AI messaging            | Django Channels/Redis upgrade; Claude API concierge (conversational listing discovery) and auto-reply assistant (covers offline businesses)               | 3-4 weeks              |
| 4 — Credit scoring v2                   | Incorporate real payment/transaction history into the scoring model; refine <code>SCORE_FACTORS</code> weighting; lending-partner integration touchpoints | 1-2 weeks              |
| 5 — DevOps hardening & launch readiness | Security review, automated backups (restic/off-site), monitoring (e.g. Uptime Kuma/Sentry), basic load testing, production runbook                        | 1-2 weeks              |
| Future — Mobile app                     | Separate discovery/spec/brainstorming cycle once the web platform and API are stable                                                                      | TBD, own project scope |

**Total Phase 2-5 estimate:** roughly **7-11 weeks** at a sustainable full-time pace (proportionally longer if part-time).

## 7. DevOps & Deployment

- **Environment:** Single Cloud VPS, mid-tier sizing (~\$50-80/mo — e.g. 4-8 vCPU / 8-16GB RAM tier)
- **Stack on the VPS (via Docker Compose):**
  - `web`: Gunicorn serving Django/DRF (Daphne/Uvicorn added in Phase 3 for Channels)
  - `db`: PostgreSQL
  - `redis`: added in Phase 3 for Channels
  - `nginx`: reverse proxy + static/media file serving + TLS via Certbot
- **CI/CD:** GitHub Actions — build on push to `main`, deploy over SSH, restart Compose stack
- **Backups:** Scheduled dump of Postgres + media volume, shipped off-site (e.g. restic to Backblaze B2/S3-compatible target) — added as part of Phase 5 hardening, though a basic cron backup should exist from Week 1 onward given real user data starts accumulating immediately
- **Secrets:** Environment variables via `.env` (not committed), consumed by Docker Compose
- **Monitoring:** Deferred to Phase 5 (Sentry for error tracking, uptime checks) — acceptable gap for a Week 1 MVP, not acceptable to carry into a real launch

## 8. Cost Considerations (Ongoing, Post-Launch)

| Item                                 | Estimate                                                                                                                                  |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| VPS hosting                          | \$50-80/mo                                                                                                                                |
| Domain + TLS                         | Domain done (theashintihumb.co); TLS via Certbot                                                                                          |
| SMS OTP                              | Per-message cost via Hubtel SMS/Africa's Talking (usage-dependent)                                                                        |
| Hubtel payment processing            | Transaction-fee based (Phase 2 — confirm rate card during integration)                                                                    |
| Claude API (AI concierge/auto-reply) | Usage-based; recommend a cheaper model tier for high-volume auto-replies and a stronger tier for the concierge, with a monthly budget cap |

|                  |                                                               |
|------------------|---------------------------------------------------------------|
| Off-site backups | ~\$30-50/mo (S3-compatible storage, low volume at this stage) |
|------------------|---------------------------------------------------------------|

## 9. Summary

Week 1 delivers a real, mock-data-free MVP: real users, real businesses, real listings, real (if payment-blind) credit scoring, all live on a production VPS. Phases 2-5 bring payments, AI-assisted real-time messaging, and a fully-informed credit score online over the following 7-11 weeks at a sustainable pace. Mobile app development is intentionally deferred to its own scoping cycle rather than bolted onto this timeline.